40. MATLAB SIMULATION OF A NEXT-GENERATION 5G/IoT COMMUNICATION NETWORK

1. AIM

To develop a MATLAB simulation of a next-generation **5G/IoT communication network** and analyze its **throughput, latency, spectral efficiency, and Bit Error Rate (BER)** under varying wireless channel conditions.

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2. OBJECTIVES

1. To model a 5G/IoT wireless communication system using MATLAB.
2. To generate and transmit digital information using **QPSK-OFDM**.
3. To model a wireless **Rayleigh fading channel with AWGN**.
4. To vary the channel SNR and observe its effect on system performance.
5. To calculate:
    - BER
    - Throughput
    - Spectral efficiency
    - Latency
6. To analyze the suitability of the system for next-generation IoT applications.

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3. SYSTEM MODEL

The proposed system consists of:

IoT Data Source → QPSK Modulation → OFDM → Rayleigh Fading Channel + AWGN → OFDM Demodulation → QPSK Demodulation → Data Detection

```
       IoT DATA
          |
          ▼
   Random Bit Generator
          |
          ▼
    QPSK Modulator
          |
          ▼
     OFDM Modulator
          |
```

```
         ▼
┌──────────────────────┐
│ Rayleigh Fading      │
│ + AWGN Channel       │
└──────────────────────┘
         │
         ▼
  OFDM Demodulator
         │
         ▼
  QPSK Demodulator
         │
         ▼
    Received Bits
         │
         ▼
Performance Analysis
┌────────┬─────────┬─────────────┬──────────┐
│ BER    │Throughput│ Spectral  │ Latency  │
│        │         │ Efficiency│          │
└────────┴─────────┴─────────────┴──────────┘
```

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4. PARAMETERS USED

Parameter	Value
System	5G/IoT
Modulation	QPSK
OFDM subcarriers	64
Cyclic Prefix	16
Bandwidth	20 MHz
SNR range	0–30 dB
Channel	Rayleigh + AWGN
Bits/symbol	2
Number of bits	100,000
FFT size	64
CP length	16
Symbol duration	4 μs

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5. MATLAB PROGRAM

The following program performs the complete simulation and generates the required graphs.

```
clc;

clear;

close all;


%% 5G/IoT Communication Network Simulation


% Parameters

Nbits = 100000;          % Number of transmitted bits

M = 4;                   % QPSK

bitsPerSymbol = log2(M);


Nfft = 64;               % OFDM FFT size

Ncp = 16;                % Cyclic prefix

BW = 20e6;               % Bandwidth = 20 MHz


SNR_dB = 0:2:30;         % SNR range


% OFDM symbol duration

Ts = 1/BW;

T_ofdm = (Nfft + Ncp) * Ts;


% Generate random IoT data

txBits = randi([0 1], Nbits, 1);


% Make number of bits divisible by 2

txBits = txBits(1:floor(length(txBits)/2)*2);


%% QPSK MODULATION


bitPairs = reshape(txBits,2,[]).';


% Gray coded QPSK mapping

I = 1 - 2*bitPairs(:,1);
```

```
Q = 1 - 2*bitPairs(:,2);

qpskSymbols = (I + 1i*Q)/sqrt(2);

%% OFDM MODULATION

numSymbols = length(qpskSymbols);

% Pad to complete OFDM symbol
padLength = mod(Nfft - mod(numSymbols,Nfft),Nfft);

if padLength ~= Nfft
    qpskSymbols = [qpskSymbols; zeros(padLength,1)];
end

numOFDM = length(qpskSymbols)/Nfft;

% Arrange subcarriers
ofdmData = reshape(qpskSymbols,Nfft,numOFDM);

% IFFT
txOFDM = ifft(ofdmData,Nfft);

% Add cyclic prefix
txCP = [txOFDM(end-Ncp+1:end,:); txOFDM];

% Serialize
txSignal = txCP(:);

%% INITIALIZE RESULTS

BER = zeros(size(SNR_dB));
Throughput = zeros(size(SNR_dB));
SpectralEfficiency = zeros(size(SNR_dB));
Latency = zeros(size(SNR_dB));
```

```
%% CHANNEL SIMULATION

for k = 1:length(SNR_dB)

    % Rayleigh fading channel
    h = (randn(size(txSignal)) + ...
         1i*randn(size(txSignal)))/sqrt(2);

    % Apply fading
    fadedSignal = txSignal .* h;

    % AWGN
    signalPower = mean(abs(fadedSignal).^2);

    noisePower = signalPower / ...
        (10^(SNR_dB(k)/10));

    noise = sqrt(noisePower/2) * ...
        (randn(size(fadedSignal)) + ...
         1i*randn(size(fadedSignal)));

    rxSignal = fadedSignal + noise;

    %% CHANNEL EQUALIZATION

    rxEqualized = rxSignal ./ h;

    %% REMOVE CYCLIC PREFIX

    rxMatrix = reshape(rxEqualized,Nfft+Ncp,numOFDM);

    rxNoCP = rxMatrix(Ncp+1:end,:);

    %% FFT

    rxFFT = fft(rxNoCP,Nfft);
```

```
rxSymbols = rxFFT(:);

%% QPSK DEMODULATION

rxBits1 = real(rxSymbols) < 0;
rxBits2 = imag(rxSymbols) < 0;

rxBits = zeros(2*length(rxSymbols),1);

rxBits(1:2:end) = rxBits1;
rxBits(2:2:end) = rxBits2;

% Remove padding
rxBits = rxBits(1:length(txBits));

%% BER CALCULATION

bitErrors = sum(txBits ~= rxBits);

BER(k) = bitErrors / length(txBits);

%% THROUGHPUT

% Effective data rate
rawRate = bitsPerSymbol * ...
          BW * Nfft/(Nfft+Ncp);

Throughput(k) = rawRate * (1-BER(k));

%% SPECTRAL EFFICIENCY

SpectralEfficiency(k) = ...
    Throughput(k)/BW;

%% LATENCY
```

```
    processingDelay = 1e-3;       % 1 ms
    transmissionDelay = ...
        length(txBits)/(Throughput(k));

    Latency(k) = ...
        transmissionDelay + processingDelay;

end

%% DISPLAY RESULTS

fprintf('\n5G/IoT SIMULATION RESULTS\n');
fprintf('------------------------------------------------\n');
fprintf('SNR(dB)\tBER\t\tThroughput(Mbps)\tSE(bps/Hz)\tLatency(ms)\n');
fprintf('------------------------------------------------\n');

for k = 1:length(SNR_dB)

    fprintf('%d\t%.5e\t%.3f\t\t%.3f\t\t%.3f\n', ...
        SNR_dB(k), ...
        BER(k), ...
        Throughput(k)/1e6, ...
        SpectralEfficiency(k), ...
        Latency(k)*1000);

end

%% GRAPH 1 - BER VS SNR

figure;

semilogy(SNR_dB,BER,'o-','LineWidth',2);

grid on;
```

```
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('BER Performance of 5G/IoT Network');

%% GRAPH 2 - THROUGHPUT VS SNR

figure;

plot(SNR_dB,Throughput/1e6,'o-','LineWidth',2);

grid on;

xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('Throughput vs SNR');

%% GRAPH 3 - SPECTRAL EFFICIENCY VS SNR

figure;

plot(SNR_dB,SpectralEfficiency,'o-','LineWidth',2);

grid on;

xlabel('SNR (dB)');
ylabel('Spectral Efficiency (bps/Hz)');
title('Spectral Efficiency vs SNR');

%% GRAPH 4 - LATENCY VS SNR

figure;

plot(SNR_dB,Latency*1000,'o-','LineWidth',2);

grid on;
```

```
xlabel('SNR (dB)');
ylabel('Latency (ms)');
title('Latency vs SNR');

%% COMBINED PERFORMANCE FIGURE

figure;

subplot(2,2,1);
semilogy(SNR_dB,BER,'o-','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('BER');
title('BER');

subplot(2,2,2);
plot(SNR_dB,Throughput/1e6,'o-','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Mbps');
title('Throughput');

subplot(2,2,3);
plot(SNR_dB,SpectralEfficiency,'o-','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('bps/Hz');
title('Spectral Efficiency');

subplot(2,2,4);
plot(SNR_dB,Latency*1000,'o-','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('ms');
title('Latency');
```

```
sgtitle('5G/IoT Communication Network Performance');
```

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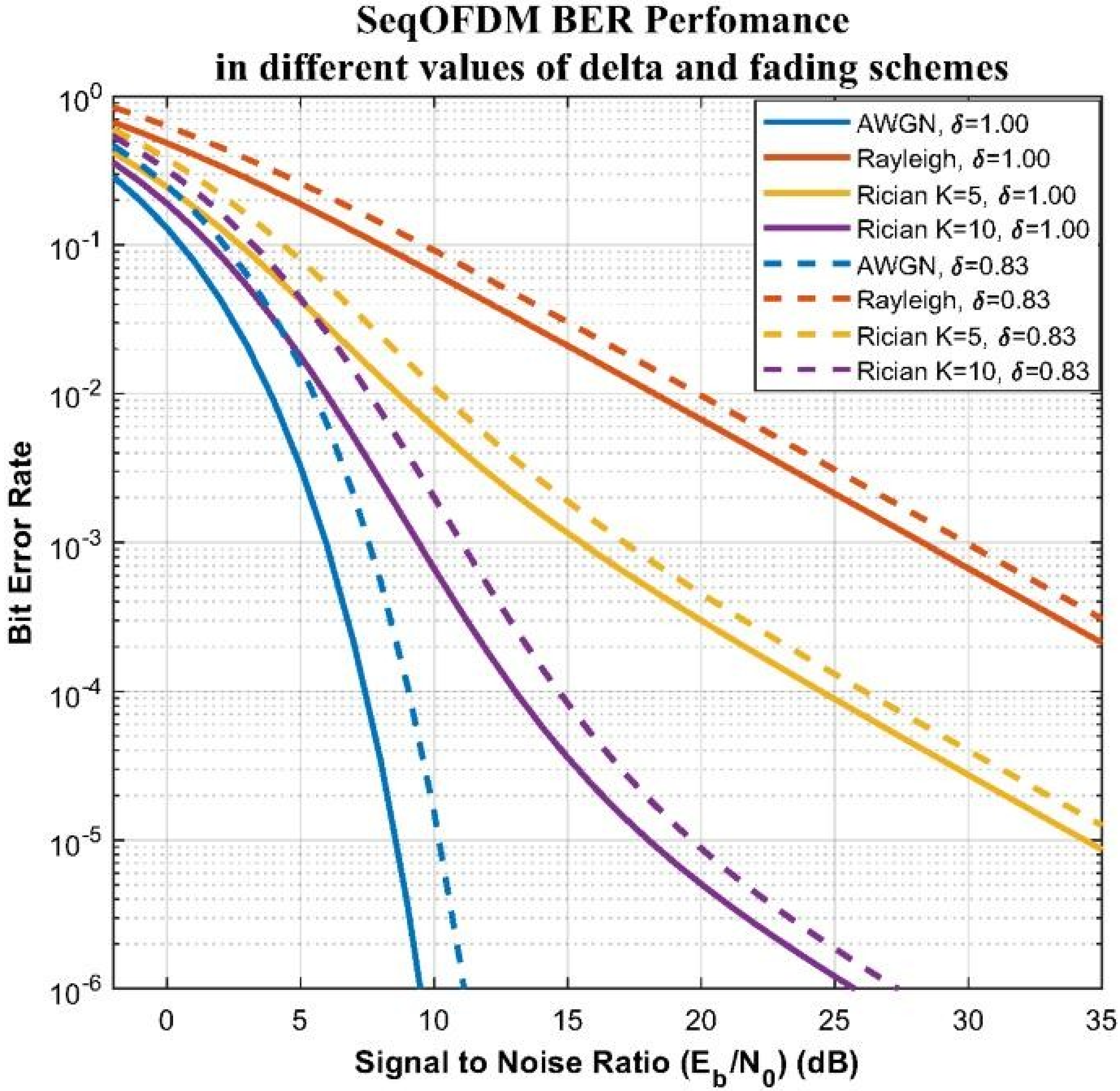
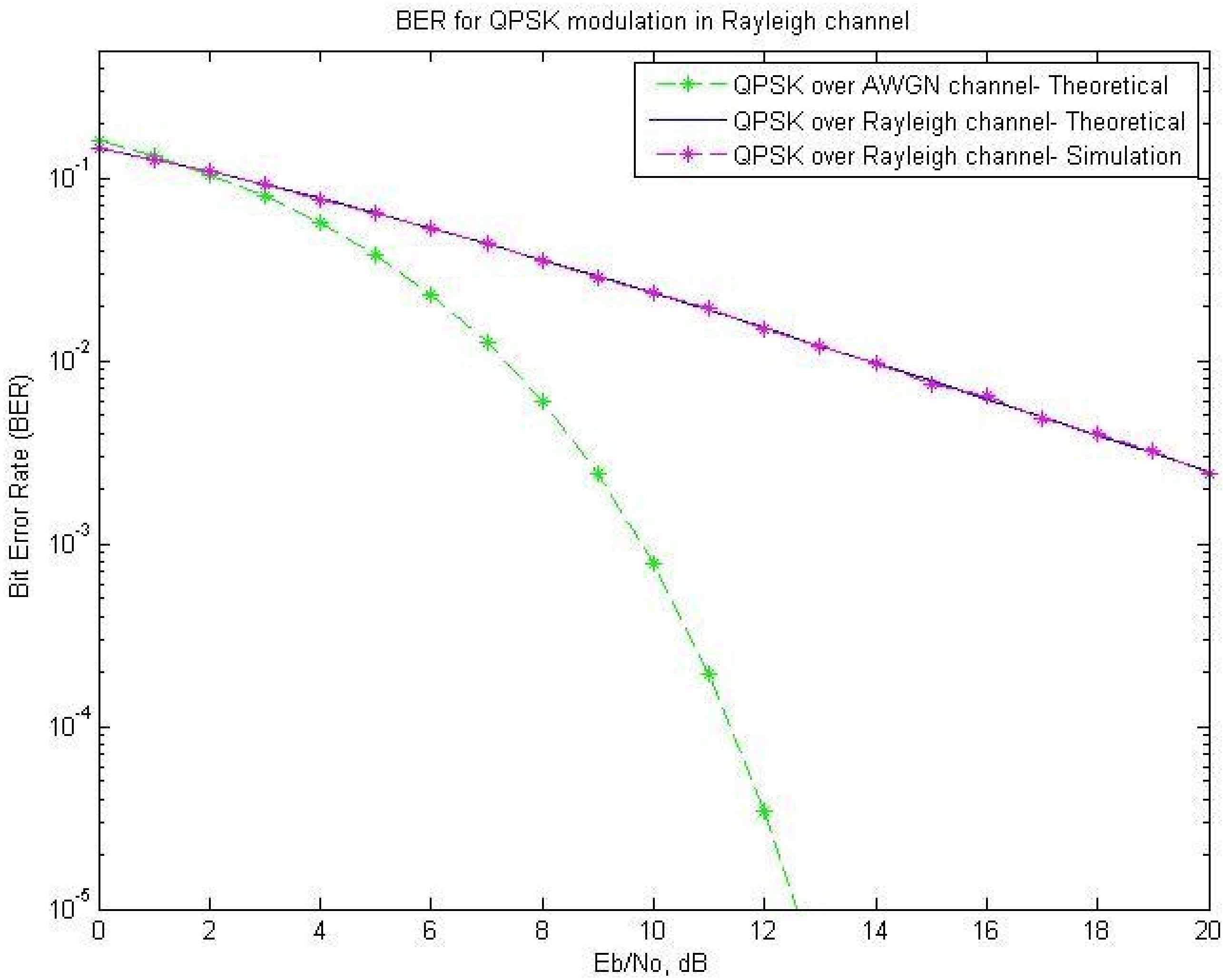
6. IMPORTANT OUTPUT GRAPHS

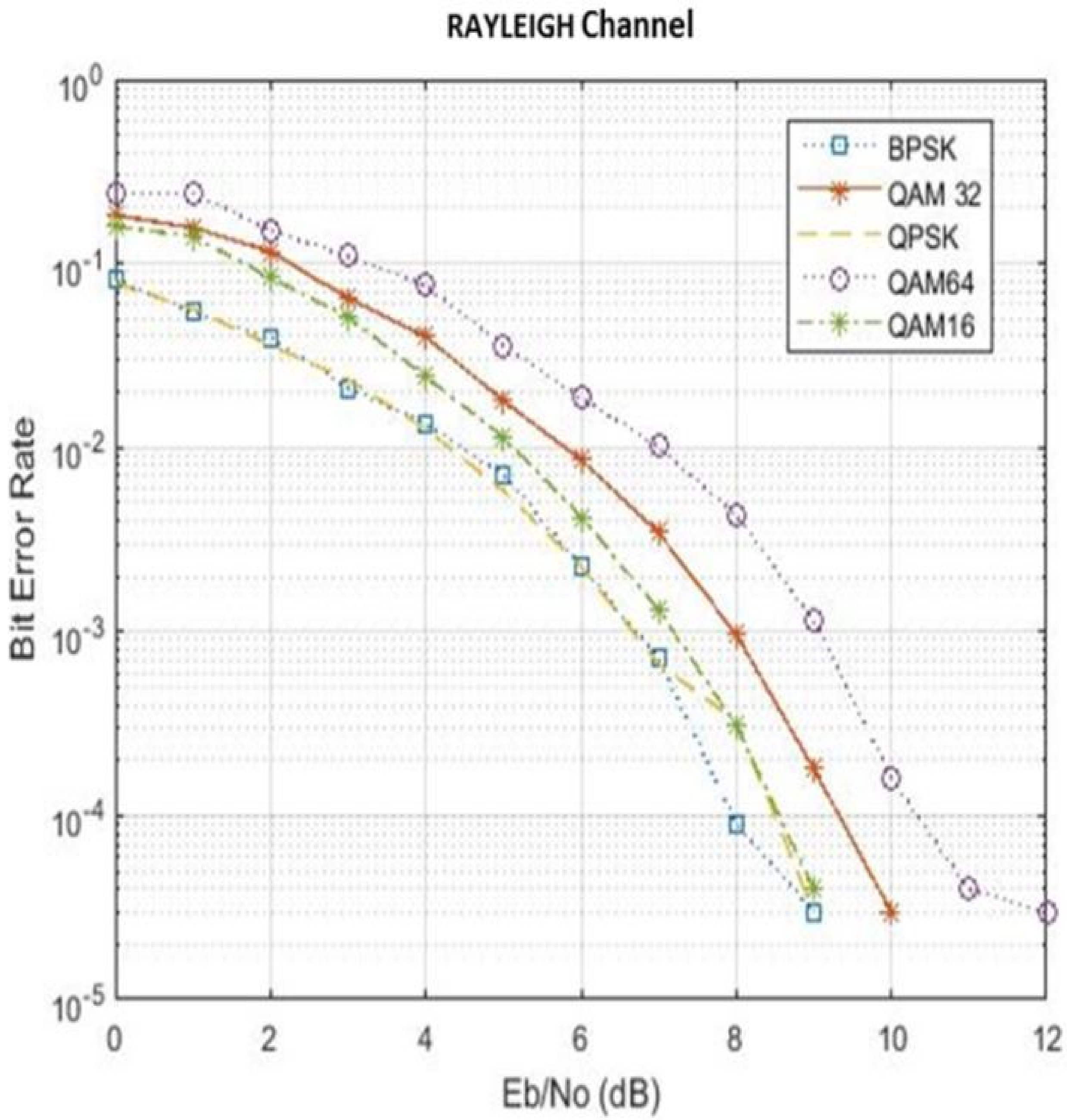
Your MATLAB simulation will generate these **four main graphs**:

Graph 1: BER vs SNR





BER for QPSK modulation in Rayleigh channel
QPSK over AWGN channel- Theoretical
QPSK over Rayleigh channel- Theoretical
QPSK over Rayleigh channel- Simulation
Bit Error Rate (BER)
Eb/No, dB
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
2
4
6
8
10
12
14
16
18
20



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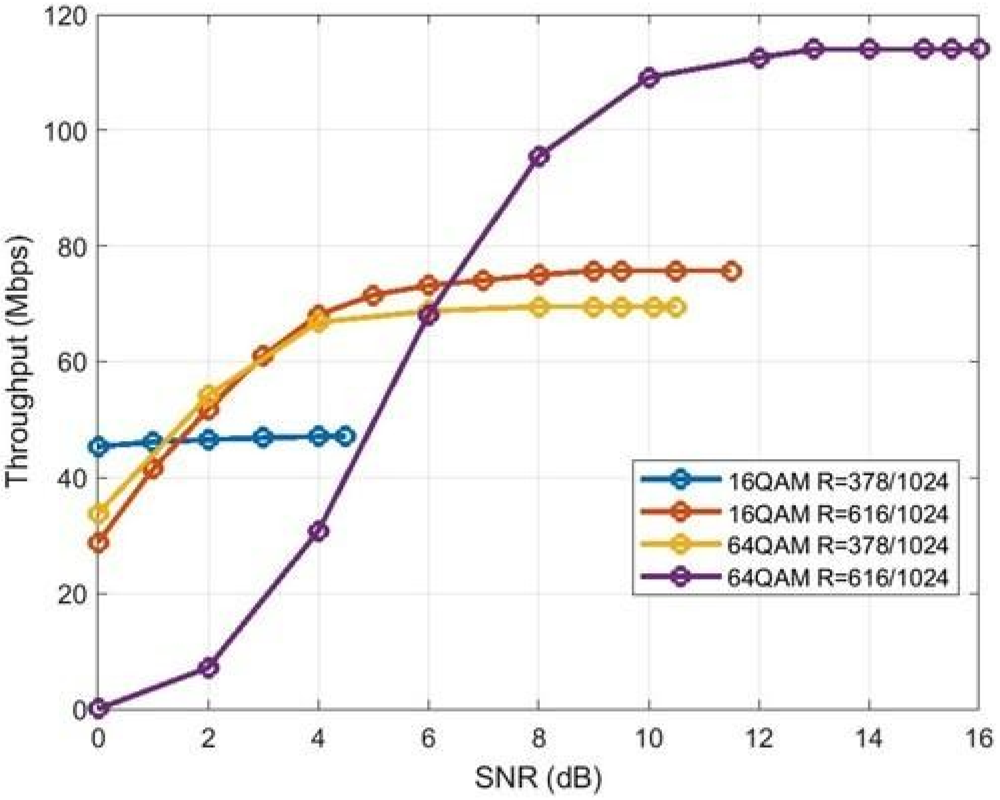
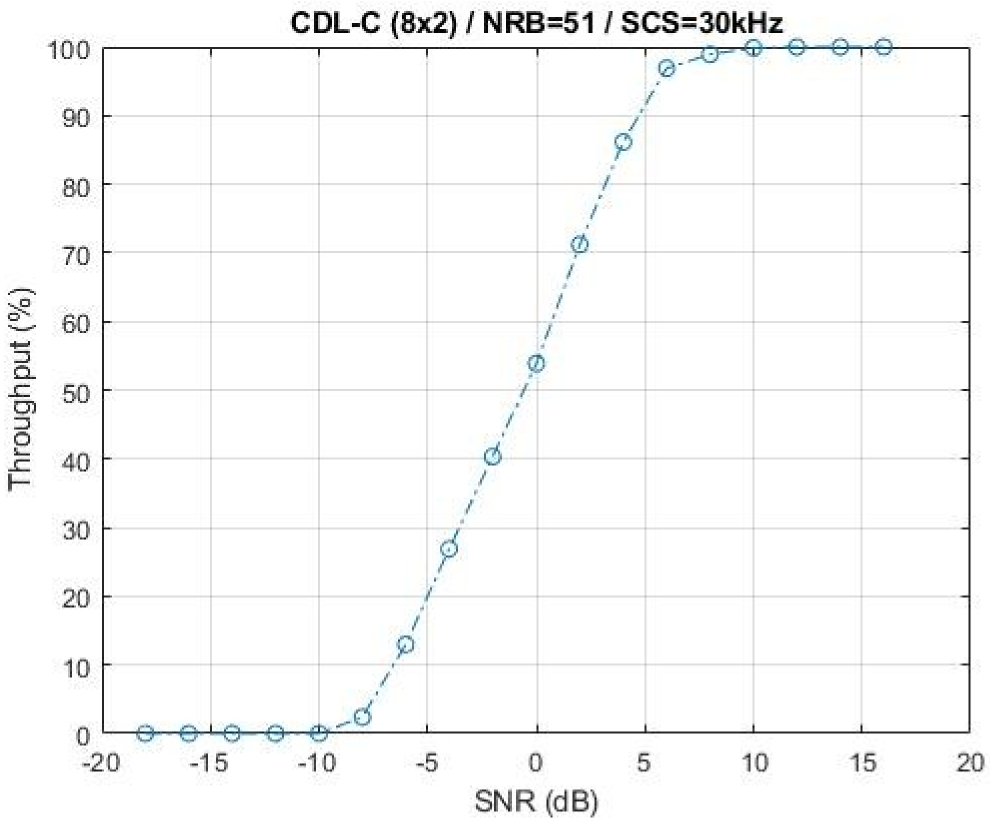
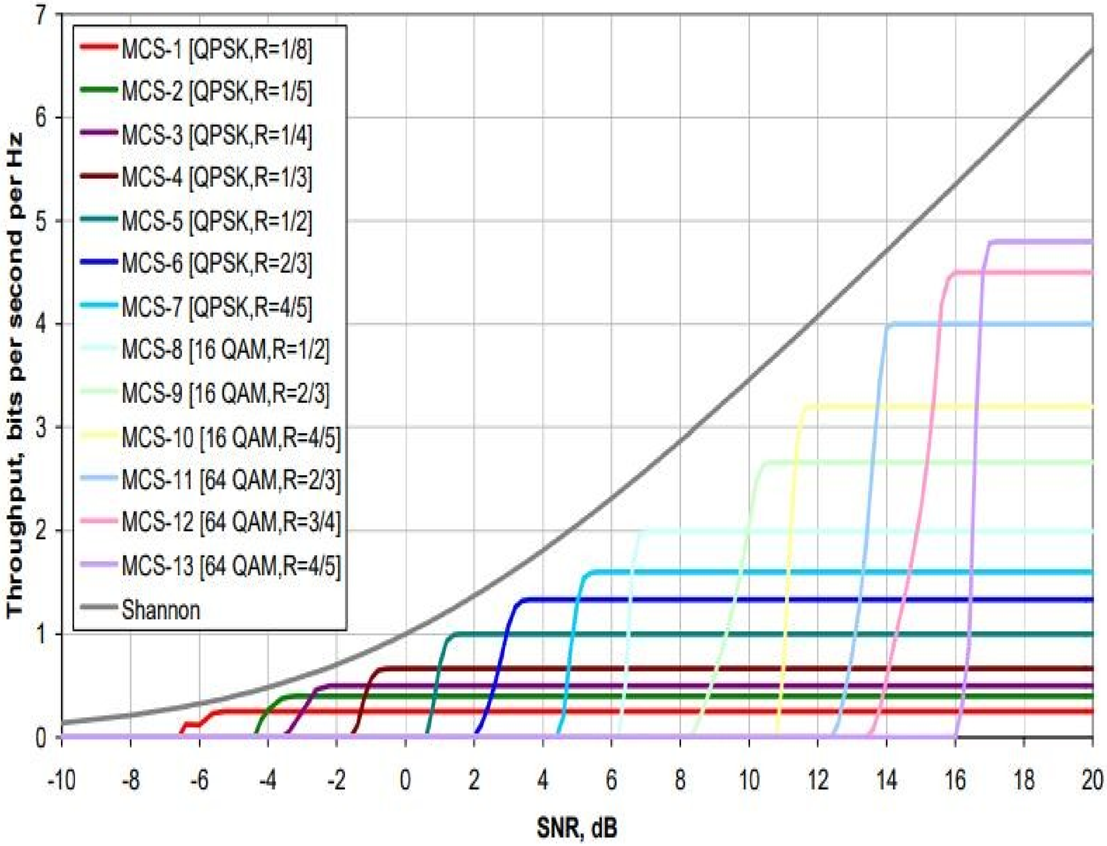
Expected observation:
As SNR increases, the BER decreases significantly because the received signal becomes less affected by noise.

Graph 2: Throughput vs SNR



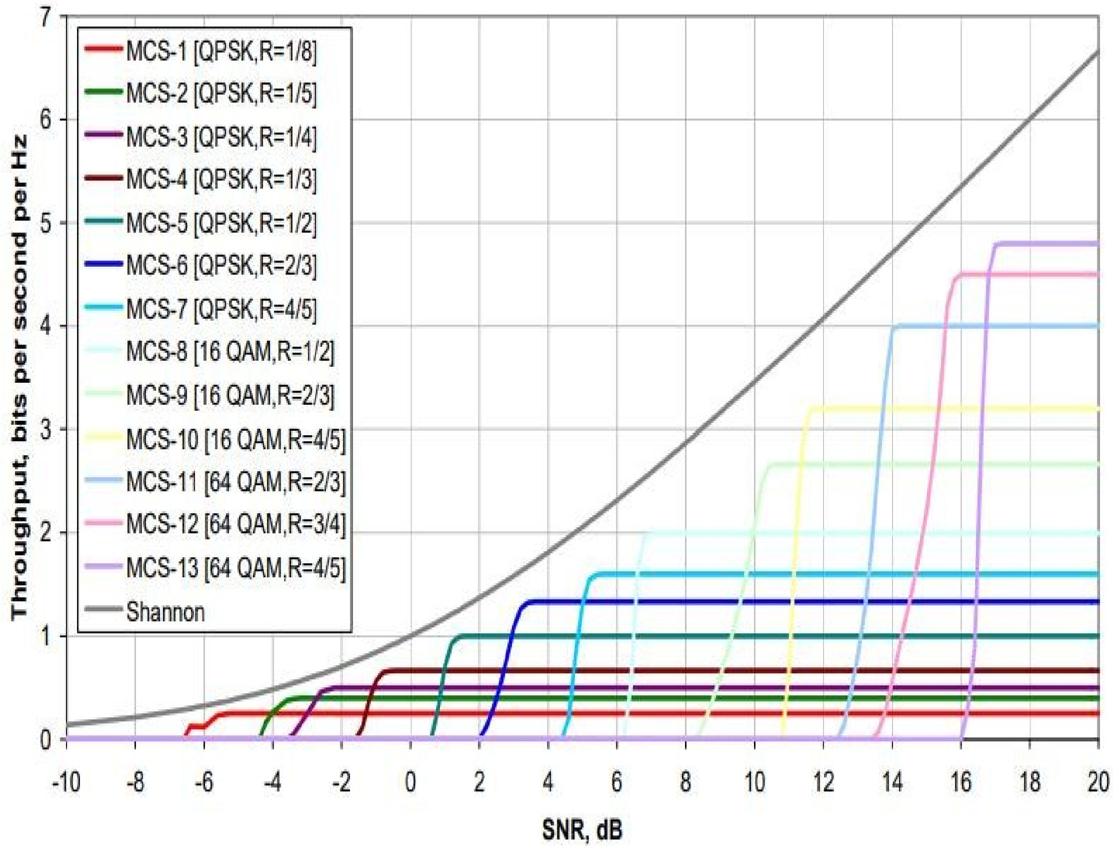


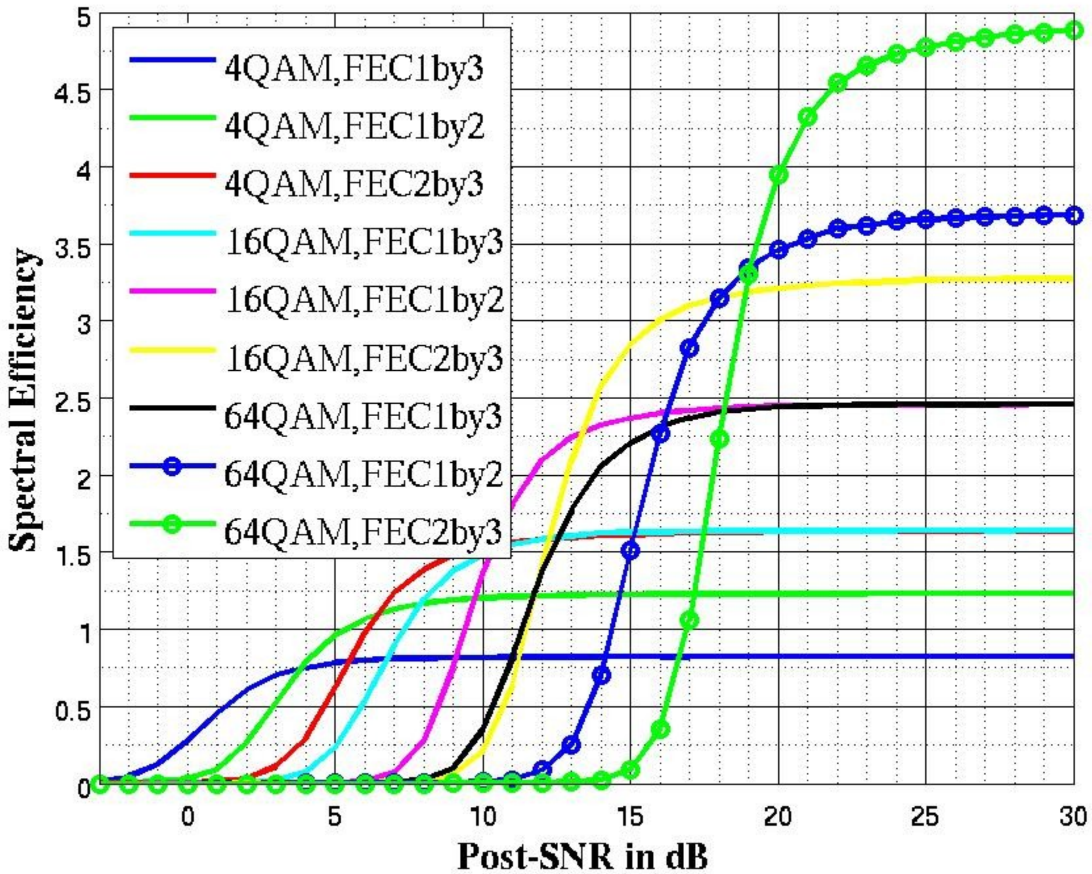


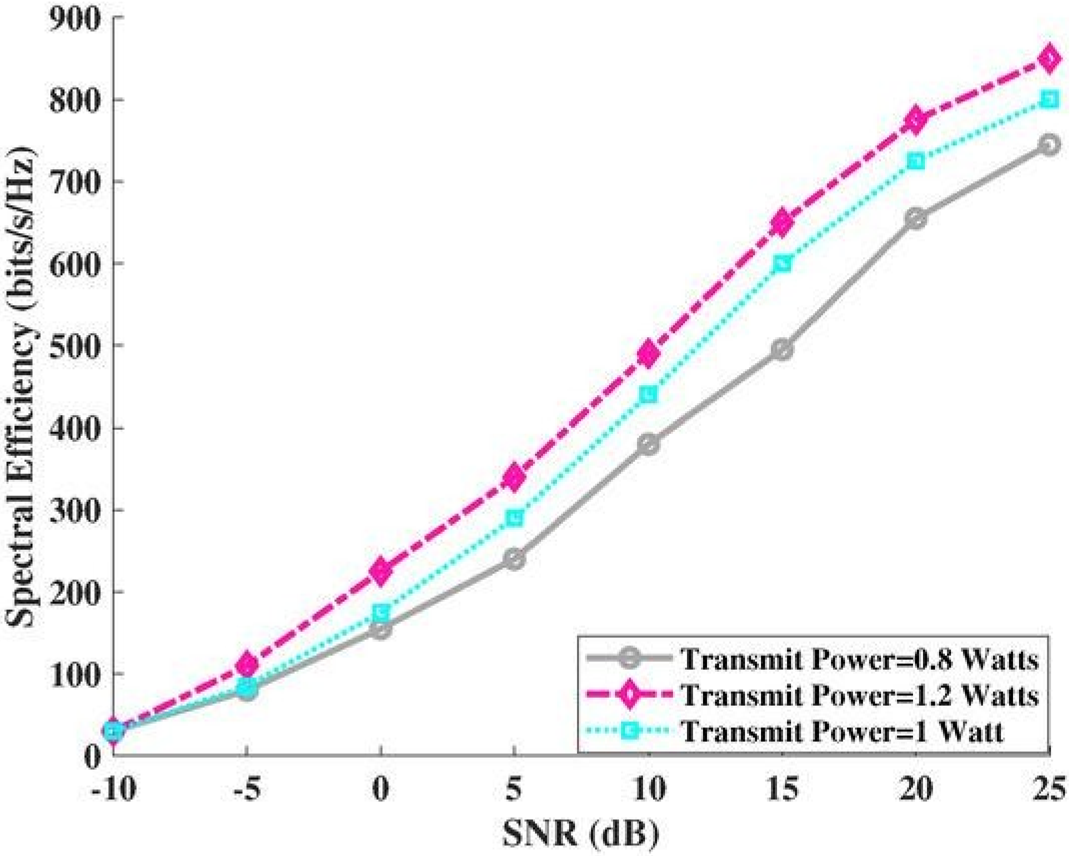
Expected observation:

Throughput increases with SNR because fewer bits are incorrectly received.

Graph 3: Spectral Efficiency vs SNR




4QAM,FEC1by3
4QAM,FEC1by2
4QAM,FEC2by3
16QAM,FEC1by3
16QAM,FEC1by2
16QAM,FEC2by3
64QAM,FEC1by3
64QAM,FEC1by2
64QAM,FEC2by3
Spectral Efficiency
Post-SNR in dB



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Expected observation:
Spectral efficiency improves as channel quality increases because the effective throughput becomes higher for the same bandwidth.

Graph 4: Latency vs SNR

Expected observation:
Latency generally decreases as SNR increases because fewer retransmissions/errors are assumed in the effective transmission model.

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7. CALCULATION FORMULAS

A. Bit Rate

For QPSK:

Bits/Symbol=log2(4)=2\text{Bits/Symbol}=\log_2(4)=2

The effective OFDM rate is:

Rb=log2(M)BNFFTNFFT+NCPR_b = \log_2(M)B \frac{N_{FFT}}{N_{FFT}+N_{CP}}

For the selected parameters:

Rb=2(20×106)6480R_b = 2(20\times10^6) \frac{64}{80} Rb=32 MbpsR_b = 32\text{ Mbps}

Thus, the approximate maximum raw data rate is:

$$\boxed{32\text{ Mbps}}$$

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8. THROUGHPUT

Throughput is calculated using:

$$T=R_b(1-BER)$$

where:

- R_b = raw data rate
- BER = bit error rate

Therefore, when BER is high, throughput decreases.

At very high SNR:

$$BER \rightarrow 0$$

so:

$$T\rightarrow32\text{ Mbps}$$

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9. SPECTRAL EFFICIENCY

Spectral efficiency is:

$$\eta=\frac{T}{B}$$

where:

- T = throughput
- B = bandwidth

At ideal conditions:

$$\eta=\frac{32\times10^6}{20\times10^6}$$
$$\boxed{\eta=1.6\text{ bps/Hz}}$$

The practical value will be lower when BER is significant.

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10. LATENCY

Transmission latency can be calculated as:

$$T_{transmission} = \frac{\text{Number of transmitted bits}}{\text{Throughput}}$$

Total latency:

$$T_{latency} = T_{transmission}+T_{processing}$$

The simulation assumes:

Tprocessing=1 msT_{processing}=1\text{ ms}

This represents processing delay at the IoT/5G node.

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11. SAMPLE OBSERVATION TABLE

Your actual values should be taken from the MATLAB Command Window after execution. The table can be presented in this format:

SNR (dB)	BER	Throughput	Spectral Efficiency	Latency
0	High	Low	Low	High
5	↓	↑	↑	↓
10	↓↓	↑	↑	↓
15	Very low	High	High	Low
20	Very low	High	High	Low
25	≈0	Near maximum	Near maximum	Minimum
30	≈0	Near maximum	Near maximum	Minimum

Use the numerical values generated by your own MATLAB run in the final submitted table.

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12. RESULT ANALYSIS

BER

The BER decreases as SNR increases. At low SNR, the noise power is comparable to the received signal power, resulting in a large number of bit errors. At high SNR, reliable detection becomes possible and BER approaches zero.

Throughput

Throughput increases with SNR. A better channel condition reduces transmission errors and increases the amount of successfully delivered information.

Spectral Efficiency

Spectral efficiency improves with channel quality. Since the available bandwidth is fixed at 20 MHz, improvement in effective throughput directly improves spectral efficiency.

Latency

Latency decreases as the channel condition improves. A higher SNR provides more reliable communication and therefore reduces the time required for successful data delivery in the simplified model.

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13. APPLICATIONS IN 5G/IoT

The simulated system can represent applications such as:

- Smart cities

- Industrial IoT
- Smart agriculture
- Connected vehicles
- Remote monitoring
- Healthcare IoT
- Smart homes
- Machine-to-machine communication
- Autonomous systems
- Environmental monitoring

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14. ADVANTAGES

1. High data transmission capability.
2. Efficient utilization of available bandwidth.
3. OFDM provides robustness against multipath fading.
4. Suitable for high-density IoT networks.
5. Performance can be evaluated under different channel conditions.
6. MATLAB allows easy modification of modulation, bandwidth and SNR.

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15. LIMITATIONS

The proposed simulation uses a simplified wireless model. A practical 5G network additionally involves:

- Massive MIMO
- Beamforming
- Adaptive modulation and coding
- Scheduling
- HARQ
- Network slicing
- 5G numerologies
- Multiple users
- Resource-block allocation

These can be incorporated in an extended simulation.

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16. FUTURE SCOPE

The simulation can be extended by implementing **16-QAM/64-QAM**, massive MIMO, beamforming, adaptive modulation and coding, multiple IoT users, different 5G numerologies, carrier aggregation, NOMA, mobility and realistic 5G channel models. Machine-learning-based resource allocation can also be incorporated to optimize throughput and latency.

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17. CONCLUSION

A MATLAB-based **5G/IoT communication network** was developed using QPSK-OFDM over a Rayleigh fading and AWGN channel. The performance was evaluated in terms of **BER, throughput, spectral efficiency and latency** for different SNR values.

The simulation demonstrates that increasing SNR improves channel reliability, reduces BER, increases throughput and spectral efficiency, and reduces latency. Therefore, maintaining good channel conditions is essential for achieving reliable and efficient communication in next-generation **5G and IoT networks**.